FFAL v3

FFAL v3 — Functional-Feature Affordance Learning on Real MuJoCo Physics

Replicable, end-to-end real-data version. No hand-typed metrics.

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Repository: `ai-papers-repo/affordance_vision/real_v3/`

Abstract

We collect a 1,020-sample affordance dataset entirely from real MuJoCo rigid-body simulations (no mock data, no hand-typed labels) and benchmark five vision backbones on the same train/val/test object split. For our ablation backbone (DINOv2 ViT-B/14 + a Functional-Invariance regularizer, denoted *FFAL*), we report test macro-AUROC = **0.8483**, output-stability FIS = **0.9610**, mean latency = **11.9 ms / image** on Apple-Silicon MPS. Crucially, **FFAL does not beat the strongest baseline (DINOv2 + plain head, 0.8632)** on macro-AUROC under this regime — and we report that openly, replacing the inflated numbers of v1/v2. We also patch the four logical holes the v2 paper acknowledged but had not actually fixed.

1. The four logical holes (v2 → v3 status)

#	v2 admission	v3 status
1	"Zero-cost" annotation is marketing	Re-framed as **Automated physics-based supervision**. Wall-clock measured: 1.2 min for 1,020 samples on this Mac mini (see §2.4).

#	v2 admission	v3 status
2	Baseline numbers were guessed	Every backbone re-run on the same images. Numbers in §3 come from `baselines_real.json`.
3	LLM context claim ungrounded	Real FFAL outputs feed a deterministic rule grader (§4). Top-1 = 0.5471, top-3 = 0.3588, κ = 0.1985.
4	FIS = 92% was latent-only	We report **both** latent FIS and output FIS for **all** five models. See §3.

2. Method

2.1 Dataset (real, not mock)

30 procedurally-generated MuJoCo objects (5 shapes × 4 surface types × randomized size/density/friction). For each object we run 34 episodes × 5 affordance probes:

- **sittable**: drop a 70 kg surrogate, measure tilt; label=1 iff tilt<12° and top area>0.015 m²
- **stackable**: drop a 0.04 m cube, measure lateral drift
- **graspable**: simulate parallel-jaw closure under 80 N, measure object slip
- **pushable**: apply 10 N for 1 s, measure displacement
- **pourable**: spawn 30 particles, count fraction retained

Total: **1020 samples**, **1020 unique 224×224 renders** with per-episode camera/light jitter.

Class balance (positive fraction):

Affordance	Positive	Negative	Pos rate
sittable	350	670	34.31%
stackable	405	615	39.71%
graspable	408	612	40.00%
pushable	578	442	56.67%
pourable	102	918	10.00%

2.2 Splits (no object leakage)

Split by `obj_id`: **714** train / **136** val / **170** test (21 / 4 / 5 objects). Test object IDs: `obj_09`, `obj_11`, `obj_12`, `obj_23`, `obj_26`.

2.3 Models compared

Backbone	Source	Trainable params	Feature dim
ResNet50	torchvision IMAGENET1K_V2	head only (~525 K)	2048
ViT-B/16	torchvision SWAG_E2E_V1	head only	768
CLIP ViT-B/32	open_clip OpenAI	head only	512
DINOv2 ViT-B/14	timm `vit_base_patch14_dinov2.lvd142m`	head only	768
FFAL (ours)	DINOv2 + same head + **FI regularizer** (λ=0.5, 4 augmented views per sample)	head only	768

2.4 Honest cost of "automated" supervision

Replacing the prior "Zero-cost" framing:

- **One-time engineering**: ~10 person-hours to design probes and MJCF templates (this entire `real_v3/` directory).
- **Per-run wall-clock** (Mac mini, MPS):
  - Data collection (1,020 samples + 1,020 renders): **1.2 min**.
  - Train 4 baseline heads + extract features: ~4 min total.
  - Train FFAL head (15 epochs, FI reg, MPS): ~12 min.
  - Stability + LLM grounding: ~1 min.
- **Marginal cost per new object archetype**: ~2.4 s of physics + 0.1 s rendering. Break-even vs. paid manual annotation kicks in around N ≈ 200 objects under standard industry rates; we run at N = 30 for proof of concept and document this honestly.

3. Results

3.1 Head-line numbers (real forward passes, real eval)

Model	Macro AUROC	Macro F1	Latent FIS	Output FIS	Latency (ms/img)
resnet50	0.8234	0.5726	0.7934	0.9406	4.9
vit_b16	0.8230	0.5995	0.9006	0.9689	41.1
clip_b32	0.8543	0.6391	0.9454	0.9626	11.9
dinov2	0.8632	0.5779	0.8823	0.9669	12.0
ffal	0.8483	0.5564	0.8882	0.9610	11.9

3.2 What this means (honest interpretation)

- **Best macro-AUROC: `dinov2` at 0.8632**. FFAL is competitive but **does not** beat plain DINOv2 on this dataset and at this training budget. We retract any prior claim of a +10 pp advantage.
- **Output FIS gap closes the v2 "math illusion" hole**: every model reports both metrics. Output FIS ranges from 0.941 (ResNet50) to 0.969 (ViT-B/16). Latent FIS is *not* a reliable proxy for downstream output stability — confirmed empirically.
- **Per-affordance AUROC** (next table) shows where each backbone wins/loses. `pourable` is near-perfect for all models because hollow-top objects are visually obvious; `graspable` is the hardest (best 0.747 by ResNet50). Reporting per-affordance honestly prevents the v2 macro-score from hiding weak categories.

3.3 Per-affordance AUROC

Model	sittable	stackable	graspable	pushable	pourable
resnet50	0.934	0.749	0.747	0.687	1.000
vit_b16	0.896	0.781	0.563	0.880	0.995
clip_b32	0.924	0.803	0.633	0.914	0.997
dinov2	0.947	0.799	0.691	0.883	0.997
ffal	0.930	0.807	0.704	0.804	0.997

4. Multi-modal LLM context — grounded, not asserted

We sampled **170** test images, formed a deterministic prompt around each FFAL prediction, and asked a *rule grader* (same rule applied to gold labels) to compare predicted top-3 "plausible uses" against the gold top-3.

- Top-1 precision: **0.5471**
- Top-3 precision: **0.3588**
- Top-3 Jaccard: **0.3118**
- Cohen's κ: **0.1985**

These numbers are modest because (a) our test set has only 5 held-out objects, and (b) the rule grader is intentionally strict (it expects exact use-string matches). The point is that the metric is **now real**: rerunning `03_llm_context.py` reproduces it exactly. Future work: replace the rule grader with a calibrated LLM judge over a larger held-out set; the hook is in place (`--call-llm` switch).

5. Limitations

- Only 30 object archetypes; macro-AUROC variance across the 5-object test set is non-trivial.

- All affordance labels come from a single physics engine (MuJoCo). Sim-to-real transfer is *not* evaluated in v3; v2 reported a 87.6% transfer figure that we cannot currently reproduce without a real-image testbed and we therefore **drop it** from the abstract.
- FI regularizer was trained for 15 epochs at batch=16 because of MPS memory; a larger budget might invert the FFAL-vs-DINOv2 ranking. We do not claim it would.
- The LLM context grader is rule-based; agreement with a calibrated language model is future work.

6. Reproducibility

Every metric in this manuscript is produced by:

```
cd ai-papers-repo/affordance_vision/real_v3
python3 scripts/01_collect_mujoco.py          # 1.2 min
python3 scripts/02_train_and_baselines.py      # ~16 min on Mac mini MPS
python3 scripts/03_llm_context.py              # <1 min
python3 scripts/04_compile_paper.py            # regenerates this PDF
```

Seed = 20260516, pinned for numpy, random, and torch. Pipeline tested on torch 2.8.0, MuJoCo 3.3.7, macOS Darwin 25.4 (arm64).

7. Appendix A — Per-metric provenance

Metric in paper	Source file	Field
All AUROC, F1, FIS, latency	`results/baselines_real.json`	results.<model>.*
LLM-context numbers	`results/llm_context_real.json`	`top1_precision`, etc.
Class balance	`data/affordances_real.jsonl`	per-row `label_*`
Object catalog	`data/object_catalog.json`	30 entries
Training history	`results/baselines_real.json`	results.<model>.train_history